

# Enshi Zhang

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## Summary

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I am a Ph.D. candidate in Computer Science at Florida International University. My research develops multimodal models at the intersection of speech and natural language processing for healthcare applications, including emotion recognition, depression detection, and longitudinal speech modeling for neurodegenerative disease. I also investigate data quality challenges in real-world health data collection, with a focus on how system design and human factors shape the reliability, completeness, and usability of health data.

## Research Interests

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My current research interests center on developing speech-language foundation models for healthcare and clinical applications. Building on my prior works, I am interested in moving beyond task-specific models toward foundation model that can support a broader range of clinical speech and language analysis tasks. I am also interested in clinical speech synthesis, particularly as a way to address the scarcity, sparsity, and imbalance of speech data in mental disorder and neurodegenerative disease studies.

## Selected Research Projects

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**Multimodal Depression Detection.** Developed a multimodal framework for depression detection from clinical interviews, addressing the challenge that models may learn shortcuts from interviewer questions rather than clinically meaningful participant behavior. Integrated participant speech, transcriptions, and interviewer prompts, and proposed adversarial learning with contextual positional encoding to improve robustness and reduce interviewer-question bias. Published at EMNLP 2025.

**Contextual Speech Emotion Recognition with LLMs.** Developed contextual utterance modeling strategies that use ASR-transcribed dialogue history and large language models to improve speech emotion recognition. Studied how conversational context and limitations of different LLMs affect emotion prediction, winning first place in the IEEE SLT GenSEC Challenge 2024; extended version published at the NeurIPS Audio Imagination Workshop 2024.

**MERSA: A Speech Emotion Recognition Benchmark.** Constructed the MERSA benchmark from speech, text, physiological data from wearable device, and survey data collected from 150 college students to support speech emotion recognition, sentiment analysis, and depression-related modeling. Developed a Transformer-based modeling framework and released both the model and the datasets publicly for affective computing and mental health AI. Published at ACL 2024.

**Data Quality and Crowdsensing.** Analyzed a longitudinal mental wellness crowdsensing study conducted from October 2022 to September 2023 to understand how app design, participant engagement, and human factors affect data quality. Derived design implications for real-world health data collection systems and longitudinal mental wellness studies. Published at ACM CSCW 2025.

## Education

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| <b>Florida International University</b> , Miami FL, USA<br>Ph.D. in Computer Science<br>Advisor: Dr. Christian Poellbauer | Aug 2021 – Present<br>(expected graduation: Fall 2026) |
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| <b>La Salle University</b> , Philadelphia PA, USA<br>Master in Business Administration | Aug 2017 – Aug 2018 |
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| <b>University of Kansas</b> , Lawrence KS, USA<br>B.S. in Mathematics & Accounting | Aug 2012 – May 2017 |
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## Research Experience

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| <b>Florida International University</b> – Miami, FL<br>Graduate Research Assistant<br>Advisor: Dr. Christian Poellbauer | Aug 2021 – Present |
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Developed independent research projects in speech-language and multimodal AI for health, including depression detection, speech emotion recognition, clinical speech synthesis, and data quality analysis.

Designed end-to-end research pipelines covering data collection, preprocessing, representation learning, model development, experimental design, ablation studies, and publication writing.

Built and maintained a Java-based mobile crowdsensing platform for longitudinal health data collection, participant monitoring, and wearable-device integration.

Mentored undergraduate and REU students on machine learning, speech-language modeling, experimental design, and paper writing.

## Publications

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- Mitigating Interviewer Bias in Multimodal Depression Detection: An Approach with Adversarial Learning and Contextual Positional Encoding** EMNLP 2025  
Enshi Zhang, Christian Poellabauer
- MADSurv: An Uncertainty-Aware Framework for Multimodal Cancer Survival Analysis** ACM-BCB 2025  
Enshi Zhang, Varun Sai Raigir, Christian Poellabauer, Adrian Kohut, John Michael Templeton, Ananda Mondal, Lexi Weintraub
- Participant Engagement and Data Quality: Lessons Learned from a Mental Wellness Crowdsensing Study** ACM SIGCHI CSCW 2025  
Enshi Zhang, Rafael Trujillo, Christian Poellabauer
- A Multi-dilated Convolution Network for Speech Emotion Recognition** Scientific Reports 2025  
 Samaneh Madanian, Olayinka Adeleye, John Michael Templeton, Talen Chen, Christian Poellabauer, Enshi Zhang, Sandra L Schneider
- Contextual Speech Emotion Recognition with Large Language Models and ASR-Based Transcriptions** NeurIPS Workshop 2024  
Enshi Zhang, Christian Poellabauer
- Improving Speech-based Emotion Recognition with Contextual Utterance Analysis and LLMs (emotion recognition challenge winner)** IEEE SLT 2024  
Enshi Zhang, Christian Poellabauer
- The MERSA Dataset and a Transformer-Based Approach for Speech Emotion Recognition** ACL 2024  
Enshi Zhang, Rafael Trujillo, Christian Poellabauer
- Predicting long-term sleep deprivation using wearable sensors and health surveys** Computers in Biology and Medicine 2023  
 Rafael Trujillo, Enshi Zhang, Christian Poellabauer
- A study on mobile crowd sensing systems for healthcare scenarios** IEEE Access 2023  
Enshi Zhang, Rafael Trujillo, Christian Poellabauer

## Awards

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- Florida International University - *Dissertation Fellowship* Mar 2026
- ACM-BCB - *Student and Young Professional Award* Oct 2025
- IEEE SLT GenSEC Challenge - *Speech Emotion Recognition Winner* Nov 2024
- AUT Workshop on Speech Analysis for Neurodegenerative Diseases and Brain Injuries - *Best presentation award* Feb 2024

## Professional Service

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**Journal Reviewer:** IEEE IoT, IEEE Sensors, ACM Computing Surveys, IEEE Transactions on Affective Computing, IEEE Transactions on Wireless Communications

**Conference Reviewer:** ACM-BCB 2024, ISMAR 2024, ICIS 2024, ICWSM 2024-2025, CHI 2025, ICASSP 2025, ACM MM 2025, ICLR 2025-2026, NeurIPS 2026

**Mentor for Capstone and REU programs at FIU:** Spring 2023, Summer 2024, Fall 2024, Spring 2025, Summer 2025, Fall 2025

## Teaching

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| Teaching Assistant - Database Management, FIU  | Summer 2025 |
| Teaching Assistant - Discrete Mathematics, FIU | Spring 2024 |
| Teaching Assistant - Data Structure, FIU       | Spring 2022 |

## Technical Skills

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**Programming:** Python, Java, SQL

**Machine Learning:** PyTorch, scikit-learn, Hugging Face Transformers, large language models, model fine-tuning